

Introduction to splines

Arystan Shokan

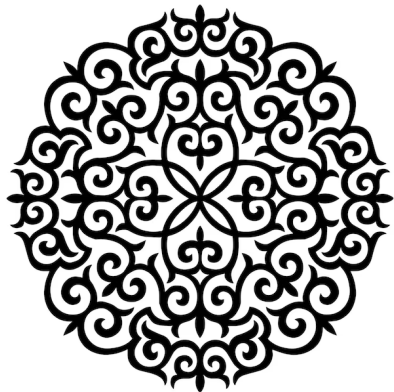
Intro to Analysis 2

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Why Splines?

- Need for computationally efficient curve representation led to piecewise low-degree polynomials X
- Splines offer *local control*: moving one knot or coefficient affects only a small portion of the curve.
- Enforce *smoothness* while using simple functions.
- Applications from CAD and graphics to data fitting demand both accuracy and performance.

Where are splines? I



Ornamental design with smooth spline curves.

Math

Scalable font.

Where are splines? II

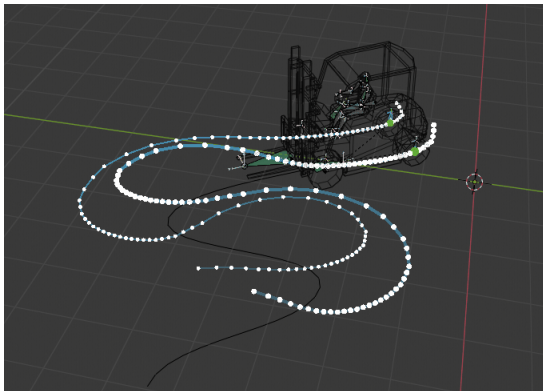


Figure: Spline-guided motion paths in 3D animation

Problem

- ① Given a set of points (Data Set) $\vec{x}_0, \vec{x}_1, \vec{x}_2, \dots, \vec{x}_m \in \mathbb{R}^n$, what is the function that passes through (or "close to") this set of points?
- ② What if we require that this function is continuous ($C^n, n \in \mathbb{N}$)?
- ③ What if it is a piecewise polynomial?

What Is a Spline?

- A *spline* is a function $S : [a, b]^n \rightarrow \mathbb{R}$ defined piecewise by polynomials.
- The partition points $a = t_0 \leq t_1 \leq \cdots \leq t_k = b$ are called *knots*.
- If each piece has degree at most n , we call S a spline of *degree* n (order $n + 1$).

Why $n = 3$ degree spline is sufficient?

- ① It is the lowest degree polynomial that can support an inflection, so we can make interesting curves
- ② It is very well behaved numerically, which means that the curves will usually be less "jumpy"

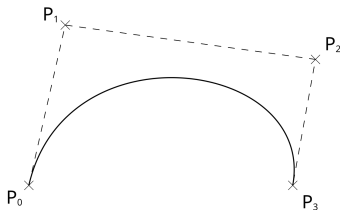
Bézier Curves

- **Definition:** For control points $\mathbf{P}_0, \dots, \mathbf{P}_n$,

$$\mathbf{B}(t) = \sum_{i=0}^n \underbrace{\binom{n}{i} t^i (1-t)^{n-i}}_{\text{Bernstein basis}} \mathbf{P}_i, \quad t \in [0, 1]$$

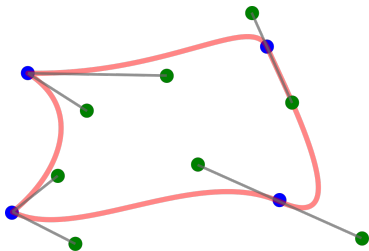
- **Key Properties:**

- Interpolates endpoints: $\mathbf{B}(0) = \mathbf{P}_0$, $\mathbf{B}(1) = \mathbf{P}_n$
- Convex hull: Curve lies within the convex hull of control points
- Global support: Moving $\mathbf{P}_i$ affects **entire curve**



Bézier Splines

- **Construction:** Connect multiple Bézier curves with continuity constraints
 - C^0 : Shared endpoint
 - C^1 : Matching first derivatives at joints
 - G^1 : Matching tangent directions (weaker)
- **Limitations:**
 - A High degree is required for complex shapes
 - No local control: Editing affects entire segments
 - Numerically unstable with many control points



Geometric Continuity (G^k)

- $G^0 = C^0$ (Positional Continuity): Ensures two curve segments meet at a common point. No gaps exist, but there may be a sharp corner.
- G^1 (Tangential Continuity): requires that the tangent directions (but not necessarily magnitudes) match at the junction. This eliminates sharp corners, creating a visually smooth transition.
- G^2 (Curvature Continuity): Ensures the curve's curvature continuity at the joint. This produces seamless transitions in how the curve bends, which is critical for high-quality surfaces.

Geometric continuity

Let $r : (a, b) \rightarrow \mathbb{R}^m$ be a curve. This curve is G^n continuous at $c \in (a, b)$ if and only if there exists a strictly increasing reparametrization

$$\sigma : (a, b) \rightarrow (a, b)$$

with $\sigma(c) = c$ so that

$$r \circ \sigma \text{ is } C^n$$

at c .

The curve is G^n continuous if it is G^n continuous at every point of its domain.

Geometric continuity vs Continuity

- ① All continuous functions are geometrically continuous.
- ② Not all geometrically continuous functions are continuous.

Example: Let $f : [0, 2] \rightarrow \mathbb{R}^2$

$$f(x) = \begin{cases} (t, t), t \in [0, 1] \\ (2t - 1, 2t - 1), t \in [1, 2] \end{cases}$$

Natural Cubic Splines

- **Definition:** $S(x)$ is C^2 on $[a, b]$, cubic on each subinterval $[x_i, x_{i+1}]$
- **Boundary Conditions:** $S''(a) = S''(b) = 0$
- **Minimal Energy:** Minimizes curvature functional

$$\mathcal{E}[f] = \int_a^b |f''(x)|^2 dx$$

- **Linear System:** For m knots, solve tridiagonal system:

$$\begin{cases} \frac{h_{i-1}}{6} M_{i-1} + \frac{h_{i-1}+h_i}{3} M_i + \frac{h_i}{6} M_{i+1} = \frac{y_{i+1}-y_i}{h_i} - \frac{y_i-y_{i-1}}{h_{i-1}} \\ M_0 = M_m = 0 \quad (\text{natural condition}) \end{cases}$$

where $h_i = x_{i+1} - x_i$, $M_i = S''(x_i)$

Error Analysis of Cubic Splines

- For $f \in C^4[a, b]$, with uniform step h , the interpolant S satisfies

$$\|f - S\|_{\infty} \leq Ch^4 \|f^{(4)}\|_{\infty}$$

.

- Derivative errors:

$$\|f' - S'\|_{\infty} \leq C_1 h^3 \|f^{(4)}\|_{\infty}, \quad \|f'' - S''\|_{\infty} \leq C_2 h^2 \|f^{(4)}\|_{\infty}$$

.

Why B-Splines? Key Advantages

Bézier Limitations Addressed

- **Local Control:** Each basis function has compact support
- **Arbitrary Continuity:** C^{p-k} at knots (order p , multiplicity k)
- **Convex Hull:** Curve lies in union of convex hulls of $p + 1$ successive points

Applications Need B-Splines

- CAD: Local editing without global recomputation
- Animation: Smooth motion with controlled derivatives
- Isogeometric Analysis: Exact geometry representation

Definition of B-Splines: Recursive Basis

- **Knot Vector:** Non-decreasing sequence $\mathbf{t} = [t_0, t_1, \dots, t_m]$
- **Cox-de Boor Recursion:**

$$B_{i,0}(t) = \begin{cases} 1 & t_i \leq t < t_{i+1} \\ 0 & \text{otherwise} \end{cases}$$

$$B_{i,p}(t) = \frac{t - t_i}{t_{i+p} - t_i} B_{i,p-1}(t) + \frac{t_{i+p+1} - t}{t_{i+p+1} - t_{i+1}} B_{i+1,p-1}(t)$$

- **Properties:**
 - Partition of unity: $\sum_i B_{i,p}(t) = 1$
 - Local support: $B_{i,p}(t) > 0$ only on $[t_i, t_{i+p+1})$
 - Differentiability: $B_{i,p} \in C^{p-1}$ at knots (for simple knots)

Non-Uniform Rational B-Splines (NURBS)

- *Definition:* NURBS extend B-splines via rational weights:

$$\mathbf{C}(u) = \frac{\sum_{i=0}^n N_{i,p}(u) w_i \mathbf{P}_i}{\sum_{i=0}^n N_{i,p}(u) w_i},$$

where $w_i > 0$ and $N_{i,p}$ are B-spline basis functions.

- *Properties:* Projective invariance, exact conic representation, local support.
- *Industry Use:* Standard in automotive and aerospace CAD for precise curve and surface modeling.

Spline-based Trajectory Planning

- *Parameterization*: Joint trajectories as piecewise polynomials with enforced C^2 continuity.
- *Optimal Control*: Minimize jerk cost

$$J = \int_0^T \|\ddot{\mathbf{q}}(t)\|^2 dt$$

subject to boundary constraints using cubic splines.

- *Kinodynamic Constraints*: Ensure velocity/acceleration limits by continuity conditions at knots.

Penalized Splines (P-splines)

- *Model*: Fit $y_i = \sum_j c_j B_j(x_i) + \epsilon_i$ with penalty

$$\lambda \int (S^{(m)}(x))^2 dx$$

on roughness.

- *Asymptotics*: Achieves optimal convergence rates for sufficiently dense knots.
- *Software*: Implemented in R via `pspline`, supports derivative evaluation.

Summary

- Splines combine local polynomial control and global smoothness.
- Continuity (C^k , G^k) tailored at knots.
- B-splines offer compact support and stability.